



Building_Energy_Modeling_Reference_Standards

BEM Reference Report: DOE, NRCan, ASHRAE 90.1-2022, and 2024 IECC

Scope

This document consolidates four PDFs into one working reference for future BEM tasks in this repo. It keeps the most reusable tables, publication dates, and interpretation notes so benchmark models, code-savings analyses, and national median benchmarking values are not mixed together.

Tables that are most reusable for prototype comparison and sanity checking are intentionally kept. Long appendices, editorial content, and tables that do not directly help with reference-building energy comparison are omitted here and should be read from the original PDFs when needed.

Unit Conversions Used Throughout:

- $1 \text{ kBtu/ft}^2 = 3.15459 \text{ kWh/m}^2$
- $1 \text{ GJ/m}^2 = 277.778 \text{ kWh/m}^2 (\sim 88.1 \text{ kBtu/ft}^2)$
- $1 \text{ ft}^2 = 0.09290304 \text{ m}^2$

Cross-Source Reading Guide

Source	Published	Geography	Stock	What the numbers are	Best use in this repo
DOE Commercial Building Benchmark Models	July 2008	U.S.	Commercial	Simulated benchmark model outputs and benchmark definitions	Baseline prototype understanding and commercial EnergyPlus sanity checks
NRCan Canadian National Median Reference Values	August 2018	Canada	Commercial and institutional	National median benchmarking values from survey/reference datasets, not simulation outputs	Real-world benchmarking and rough calibration checks
ANSI/ASHRAE/IES Standard 90.1-2022: Energy Savings Analysis	February 2024	U.S.	Commercial	Weighted simulation results comparing Standard 90.1-2019 vs 90.1-2022	Code delta expectations for commercial models
Energy Savings Analysis: 2024 IECC for Residential Buildings	December 2024	U.S.	Residential	Weighted simulation results comparing 2021 IECC vs 2024 IECC	Code delta expectations for residential models

1. DOE Commercial Building Benchmark Models (July 2008)

Item	Value
Full title	DOE Commercial Building Benchmark Models
Publication date	July 2008
Authors	P. Torcellini, M. Deru, B. Griffith, K. Benne (NREL); M. Halverson, D. Winiarski (PNNL); D.B. Crawley (DOE)
Coverage	U.S. commercial buildings
Model family	15 benchmark building types, 16 climate locations, 3 vintages (pre-1980, post-1980, new construction)
Simulation engine	EnergyPlus Version 2.2
Baseline standard	ASHRAE 90.1-2004 (new construction), 90.1-1989 (post-1980)
Most useful extracted tables	Table 1 benchmark characteristics, Table 3 climate locations, Table 8 new-construction site annual EUI
Interpretation	This is a benchmark-model definition paper. The EUI values are simulated benchmark outputs, not measured stock medians.

1.1 Benchmark Building Characteristics

Benchmark building	Floor area (ft2)	Floor area (m2)	Floors	Aspect ratio / shape	2003 CBECS area (ft2)
Large Office	460,240	42,758	12	1.5	228,725
Medium Office	53,630	4,982	3	1.5	13,842
Small Office	5,500	511	1	1.5	5,579
Warehouse	52,050	4,836	1	1.5	21,603
Stand-Alone Retail	41,790	3,882	1	1.3	10,028
Strip Mall	24,010	2,231	1	3 (per store)	23,223
Primary School	73,960	6,872	2	E-shape	26,828
Secondary School	210,890	19,594	3	E-shape	37,024
Supermarket	45,000	4,181	1	1.3	8,314
Fast Food	2,500	232	1	1	3,345
Restaurant	5,500	511	1	1	6,585
Hospital	201,250	18,698	5	1.3	241,416
Outpatient Health Care	10,000	929	2	1.5	10,409
Small Hotel	21,080	1,958	2	L-shape	14,990
Large Hotel	100,820	9,367	6	3.8 (1st flr) / 5.1	97,102

1.2 Climate Location Mapping

Climate zone	Representative city	Weather file location
1A	Miami, FL	Miami, FL
2A	Houston, TX	Houston, TX
2B	Phoenix, AZ	Phoenix, AZ
3A	Atlanta, GA	Atlanta, GA
3B	Los Angeles, CA	Los Angeles, CA
3B	Las Vegas, NV	Las Vegas, NV
3C	San Francisco, CA	San Francisco, CA
4A	Baltimore, MD	Baltimore, MD
4B	Albuquerque, NM	Albuquerque, NM
4C	Seattle, WA	Seattle, WA
5A	Chicago, IL	Chicago-O'Hare, IL
5B	Denver, CO	Boulder, CO
6A	Minneapolis, MN	Minneapolis, MN
6B	Helena, MT	Helena, MT
7	Duluth, MN	Duluth, MN
8	Fairbanks, AK	Fairbanks, AK

1.3 HVAC Equipment for New Construction and Post-1980

Building Type	Heating	Cooling	Air Distribution
Large Office	Boiler	Chiller	MZ VAV
Medium Office	Furnace	PACU	MZ VAV
Small Office	Furnace	PACU	SZ CAV
Warehouse	Furnace	PACU	SZ CAV
Stand-Alone Retail	Furnace	PACU	SZ CAV
Strip Mall	Furnace	PACU	SZ CAV
Primary School	Boiler	PACU	CAV
Secondary School	Boiler	Chiller	MZ VAV
Supermarket	Furnace	PACU	CAV
Fast Food	Furnace	PACU	SZ CAV
Restaurant	Furnace	PACU	SZ CAV
Hospital	Boiler	Chiller	FCU, CAV and VAV
Outpatient Health Care	Furnace	PACU	CAV
Small Hotel (Motel)	ISH	IRAC	SZ CAV
Large Hotel	Boiler	Chiller	FCU and VAV

PACU = Packaged Air Conditioning Unit; ISH = Individual Space Heater; IRAC = Individual Room Air Conditioner; SZ = Single Zone; MZ = Multizone; CAV = Constant Volume; VAV = Variable Air Volume; FCU = Fan Coil Units

1.4 New-Construction Site Annual EUI

Units below are from DOE Table 8: 1000 Btu/ft², effectively kBtu/ft²-yr.

Warm and Mixed Climates

Building type	1A Miami	2A Houston	2B Phoenix	3A Atlanta	3B Los Angeles	3B Las Vegas	3C San Francisco	4A Baltimore	4B Albuquerque	4C Seattle
Large Office	49.5	51.5	51.1	49.2	42.6	53.3	42.6	52.6	51.4	44.8
Medium Office	48.3	49.0	50.6	47.5	41.3	47.2	42.0	50.9	46.5	44.7
Small Office	36.7	33.6	35.2	29.1	26.1	30.1	26.1	30.5	28.9	28.2
Warehouse	18.2	21.4	21.8	26.6	17.9	24.4	25.1	34.7	29.6	32.9
Stand-Alone Retail	37.4	33.7	33.8	32.0	30.1	33.1	29.8	32.6	32.1	30.3
Strip Mall	38.1	36.2	37.6	35.2	29.2	36.1	32.3	38.5	36.3	35.4
Primary School	94.5	93.0	91.3	85.5	72.3	82.5	91.8	92.9	82.5	79.5
Secondary School	86.5	85.0	92.3	78.8	66.4	83.0	84.7	85.1	81.7	71.0
Supermarket	180.9	185.4	171.7	180.8	158.4	163.6	167.3	191.0	171.8	181.1
Fast Food	454.5	464.4	431.7	476.2	365.9	439.5	400.3	552.5	483.1	486.1
Restaurant	260.4	266.4	249.0	274.7	204.0	253.7	228.7	324.5	278.4	283.2
Hospital	126.1	128.7	132.5	121.9	99.8	122.6	107.8	132.6	123.9	113.6
Outpatient Health Care	40.2	37.0	37.7	34.4	29.7	35.6	29.1	36.8	34.4	32.5
Small Hotel	60.7	60.3	59.4	60.6	58.0	59.1	58.4	63.4	63.5	61.0
Large Hotel	68.3	72.9	75.9	64.7	50.3	63.8	57.6	73.7	65.3	65.5

Cold Climates and 2003 CBECS Comparison

Building type	5A Chicago	5B Denver	6A Minneapolis	6B Helena	7 Duluth	8 Fairbanks	2003 CBECS avg
Large Office	51.5	52.7	54.7	56.7	55.9	68.0	98.6
Medium Office	50.1	47.6	53.9	50.6	54.8	66.3	93.8
Small Office	31.7	29.4	34.6	32.1	35.3	47.3	79.9
Warehouse	42.0	36.9	53.3	47.9	58.5	87.6	47.8
Stand-Alone Retail	32.8	32.0	34.8	33.3	35.1	41.1	70.1
Strip Mall	40.9	37.9	46.6	43.5	48.5	63.4	110.0
Primary School	100.0	88.0	115.2	99.9	121.8	165.5	68.3
Secondary School	88.9	84.2	100.6	92.5	105.2	144.0	79.9
Supermarket	200.1	181.3	213.6	197.0	222.1	264.1	214.4
Fast Food	621.6	543.5	709.1	626.1	781.3	1043.6	450.9
Restaurant	366.4	314.6	421.0	367.0	464.0	633.1	231.2
Hospital	140.6	130.3	154.2	140.2	161.3	206.8	249.2
Outpatient Health Care	37.5	34.1	41.5	37.3	41.2	56.0	94.6
Small Hotel	67.2	66.2	73.2	70.6	77.6	92.5	74.9
Large Hotel	81.1	70.3	90.8	80.3	95.8	129.2	110.0

CZ6A (Minneapolis) Summary in kWh/m2

Building Type	Site EUI (kBtu/ft ²)	Site EUI (kWh/m ²)
Large Office	54.7	172.5
Medium Office	53.9	170.0
Small Office	34.6	109.1
Warehouse	53.3	168.1
Stand-Alone Retail	34.8	109.8
Strip Mall	46.6	147.0
Primary School	115.2	363.4
Secondary School	100.6	317.3
Supermarket	213.6	673.6
Fast Food	709.1	2,237.0
Restaurant	421.0	1,328.0
Hospital	154.2	486.4
Outpatient Health Care	41.5	130.9
Small Hotel	73.2	230.9
Large Hotel	90.8	286.3

1.5 Practical Reading Notes

- DOE 2008 is the oldest source in this set, but it is still important because it defines the benchmark-model logic that later commercial prototype work builds on.
 - The paper includes three vintages, but the most reusable comparative energy table in the paper is Table 8 for new construction.
 - CBECS 2003 averages include all building vintages (not just new construction), which explains why some CBECS values are higher than the new-construction benchmarks.
 - For repo work, this source is strongest when you need archetype shape, size, HVAC type, and climate sensitivity, not current code compliance targets.
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2. NRCan Canadian National Median Reference Values (August 2018)

Item	Value
Full title	Canadian National Median Reference Values for All Portfolio Manager Property Types
Publication date	August 2018
Coverage	Canada, commercial and institutional property types
Base datasets	SCIEU 2014, SECA 2014, CIBEUS 2000, EPA data for data centres, AWWA data for water/wastewater
Interpretation	This is a benchmarking reference sheet for national medians. Except for data-centre PUE, these values are not simulation outputs.
Best use	Post-simulation benchmarking, calibration sanity checks, and Canadian-market context

2.1 Dataset Basis Called Out by NRCan

Dataset	Role in the NRCan sheet
SCIEU 2014	Main source for many commercial and institutional medians
SECA 2014	Arena-related medians
CIBEUS 2000	Older legacy categories and gaps not covered well in SCIEU
EPA Data Center reference	Used for data-centre PUE where Canadian equivalent data was not available
AWWA references	Used for drinking-water and wastewater treatment facilities

2.2 Selected Archetypes Relevant to BEM Work

NRCan reports source and site EUI in GJ/m^2 . Approximate converted values use $1 \text{ GJ/m}^2 \approx 88.1 \text{ kBtu/ft}^2$ and $1 \text{ GJ/m}^2 = 277.778 \text{ kWh/m}^2$.

Property type	Source EUI (GJ/m2)	Site EUI (GJ/m2)	Site EUI (kWh/m2)	Approx site (kBtu/ft2)	Peer-group reference
Office*	1.55	0.99	275.0	87.2	SCIEU - Offices
Medical Office*	1.48	0.98	272.2	86.3	SCIEU - Medical Offices
K-12 School*	1.07	0.75	208.3	66.0	SCIEU - Primary and High Schools
College/University	1.44	1.01	280.6	88.9	SCIEU - Other (Post Secondary)
Hospital (General Medical & Surgical)*	3.09	2.22	616.7	195.5	SCIEU - Hospitals
Urgent Care/Clinic/Other Outpatient	1.47	1.02	283.3	89.8	CIBEUS - Out Patient Care
Hotel	1.37	0.88	244.4	77.5	SCIEU - Hotel and Motel
Multifamily Housing	N/A	N/A	N/A	N/A	N/A in source sheet
Supermarket/Grocery Store*	3.50	1.91	530.6	168.2	SCIEU - Food and Beverage Sales
Convenience Store*	3.50	1.91	530.6	168.2	SCIEU - Food and Beverage Sales
Fast Food Restaurant	4.12	3.17	880.6	279.1	SCIEU - Other (Restaurants)
Restaurant	4.12	3.17	880.6	279.1	SCIEU - Other (Restaurants)
Retail Store	1.60	1.13	313.9	99.5	SCIEU - Retail (Standalone)
Strip Mall	0.84	0.74	205.6	65.2	SCIEU - Retail (Strip Malls)
Enclosed Mall	4.73	4.51	1,252.8	397.1	SCIEU - Retail (Enclosed Malls)
Warehouse/Distribution Centre	1.02	0.63	175.0	55.5	SCIEU - Non-refrigerated Warehouse
Refrigerated Warehouse	2.38	1.33	369.4	117.1	SCIEU - Refrigerated Warehouse
Ice/Curling Rink*	1.76	1.15	319.4	101.3	SECA - Arenas
Indoor Arena	2.49	1.65	458.3	145.3	SECA - Large Facilities (limited data)
Data Centre (PUE, not EUI)	1.82	1.82	--	160.3	EPA - Data Center

2.3 Full NRCan Reference by Category

Office and Financial

Primary Function	Site EUI (GJ/m2)	Site EUI (kWh/m2)	Source EUI (GJ/m2)	Reference Data
Bank Branch / Financial Office	1.55	430.6	0.99	SCIEU - Offices
Office	1.55	430.6	0.99	SCIEU - Offices
Medical Office	1.48	411.1	0.98	SCIEU - Medical Offices
Veterinary Office	1.47	408.3	1.02	CIBEUS - Out Patient Care

Education

Primary Function	Site EUI (GJ/m2)	Site EUI (kWh/m2)	Source EUI (GJ/m2)	Reference Data
Adult Education	1.41	391.7	1.18	CIBEUS - Adult Education
College/University	1.44	400.0	1.01	SCIEU - Other (Post Secondary)
K-12 School	1.07	297.2	0.75	SCIEU - Primary and High Schools
Pre-school/Daycare	1.37	380.6	0.83	SCIEU - Other
Vocational School	1.41	391.7	1.18	CIBEUS - Adult Education
Other - Education	1.24	344.4	0.92	CIBEUS - Education

Healthcare

Primary Function	Site EUI (GJ/m2)	Site EUI (kWh/m2)	Source EUI (GJ/m2)	Reference Data
Hospital (General Medical & Surgical)	3.09	858.3	2.22	SCIEU - Hospitals
Other/Specialty Hospital	3.09	858.3	2.22	SCIEU - Hospitals
Ambulatory Surgical Centre	1.47	408.3	1.02	CIBEUS - Out Patient Care
Outpatient Rehab/Physical Therapy	1.47	408.3	1.02	CIBEUS - Out Patient Care
Residential Care Facility	1.47	408.3	1.04	SCIEU - Nursing Care
Senior Care Community	1.47	408.3	1.04	SCIEU - Nursing Care
Urgent Care/Clinic/Other Outpatient	1.47	408.3	1.02	CIBEUS - Out Patient Care

Lodging and Residential

Primary Function	Site EUI (GJ/m2)	Site EUI (kWh/m2)	Source EUI (GJ/m2)	Reference Data
Barracks	1.98	550.0	1.45	CIBEUS - Accommodation
Hotel	1.37	380.6	0.88	SCIEU - Hotel and Motel
Multifamily Housing	N/A	N/A	N/A	N/A
Prison/Incarceration	1.67	463.9	1.28	CIBEUS - Public Order/Safety
Residence Hall/Dormitory	1.98	550.0	1.45	CIBEUS - Accommodation
Other - Lodging/Residential	1.37	380.6	0.88	SCIEU - Hotel and Motel

Retail

Primary Function	Site EUI (GJ/m2)	Site EUI (kWh/m2)	Source EUI (GJ/m2)	Reference Data
Retail Store	1.60	444.4	1.13	SCIEU - Retail (Standalone)
Automobile Dealership	1.60	444.4	1.13	SCIEU - Retail (Standalone)
Wholesale Club/Supercentre	1.60	444.4	1.13	SCIEU - Retail (Standalone)
Strip Mall	0.84	233.3	0.74	SCIEU - Retail (Strip Malls)
Enclosed Mall	4.73	1,313.9	4.51	SCIEU - Retail (Enclosed Malls)

Food Sales and Food Service

Primary Function	Site EUI (GJ/m2)	Site EUI (kWh/m2)	Source EUI (GJ/m2)	Reference Data
Convenience Store (with/without gas)	3.50	972.2	1.91	SCIEU - Food & Beverage
Supermarket/Grocery Store	3.50	972.2	1.91	SCIEU - Food & Beverage
Fast Food Restaurant	4.12	1,144.4	3.17	SCIEU - Other (Restaurants)
Restaurant/Bar	4.12	1,144.4	3.17	SCIEU - Other (Restaurants)

Warehouse and Storage

Primary Function	Site EUI (GJ/m2)	Site EUI (kWh/m2)	Source EUI (GJ/m2)	Reference Data
Self-Storage Facility	1.02	283.3	0.63	SCIEU - Non-refrig. Warehouse
Distribution Centre	1.02	283.3	0.63	SCIEU - Non-refrig. Warehouse
Non-Refrigerated Warehouse	1.02	283.3	0.63	SCIEU - Non-refrig. Warehouse
Refrigerated Warehouse	2.38	661.1	1.33	SCIEU - Refrig. Warehouse

Entertainment, Public Assembly, Public Services, and Other

Primary Function	Site EUI (GJ/m2)	Site EUI (kWh/m2)	Source EUI (GJ/m2)	Reference Data
Convention Centre / Museum / Performing Arts	2.43	675.0	1.74	CIBEUS - Public Assembly
Movie Theatre / Bar / Nightclub / Casino	1.59	441.7	0.93	CIBEUS - Casino, Movie, Night Club
Ice/Curling Rink	1.76	488.9	1.15	SECA - Arenas
Fitness Centre/Health Club/Gym	1.89	525.0	1.51	CIBEUS - Fitness Centre
Indoor Arena	2.49	691.7	1.65	SECA - Large Facilities
Courthouse / Police Station / Prison	1.67	463.9	1.28	CIBEUS - Public Order/Safety
Library / Social Hall	2.43	675.0	1.74	CIBEUS - Public Assembly
Fire Station	1.59	441.7	1.23	CIBEUS - Municipal Admin.
Worship Facility	0.84	233.3	0.69	SCIEU - Place of Worship
Laboratory / Mixed Use / Other	1.42	394.4	0.98	SCIEU - Other
Data Centre (PUE, not EUI)	1.82	--	1.82	EPA - Data Center

2.4 Practical Reading Notes

- NRCAN is the only Canadian source in this set and the only one built primarily from survey/reference datasets rather than simulation runs.
- It is useful for benchmarking output ranges, but it should not be treated as a direct replacement for DOE or ASHRAE prototype simulations.
- Canadian climate is generally colder than U.S. averages, which contributes to higher heating loads.
- Multifamily Housing has no NRCAN median value (N/A).
- Data-centre values in the source are PUE-based, not true building EUI, so they should be handled separately from other building types.

3. ANSI/ASHRAE/IES Standard 90.1-2022: Energy Savings Analysis (February 2024)

Item	Value
Full title	ANSI/ASHRAE/IES Standard 90.1-2022: Energy Savings Analysis
Publication date	February 2024
Coverage	U.S. commercial buildings
Model family	16 commercial prototype buildings with national floor-area weights
Simulation engine	EnergyPlus (current at time of study)
Main comparison	Standard 90.1-2019 vs Standard 90.1-2022
Key distinction	Reports both "Gross" (total energy consumed) and "Net" (minus on-site renewable generation)
Interpretation	This is a weighted code-savings simulation study, not a benchmark-model-definition paper and not a measured stock median dataset.

3.1 National Weighted Summary

Metric	90.1-2019	90.1-2022 (Net)	Net Savings
Site EUI (kBtu/ft2-yr)	47.8	41.1	14.0%
Source EUI (kBtu/ft2-yr)	108.5	92.5	14.7%
Energy Cost (\$/ft2-yr)	1.35	1.15	14.8%
Carbon Emissions (tons/kft2-yr)	7.5	6.4	14.7%

3.2 Building-Type Comparison (Net Energy, including on-site renewables)

Category	Prototype	Weight	2019 site	2022 site	Site savings	2019 source	2022 source	Source savings
Office	Small Office	3.8%	28.2	25.3	10.3%	78.3	70.1	10.5%
Office	Medium Office	5.0%	30.8	24.8	19.5%	79.6	62.8	21.1%
Office	Large Office	3.9%	53.9	49.2	8.7%	147.6	134.8	8.7%
Retail	Standalone Retail	10.9%	46.8	36.7	21.6%	103.7	80.4	22.5%
Retail	Strip Mall	3.7%	50.2	37.8	24.7%	121.4	92.0	24.2%
Education	Primary School	4.8%	43.9	38.9	11.4%	101.6	89.4	12.0%
Education	Secondary School	10.9%	39.1	33.7	13.8%	94.0	80.2	14.7%
Healthcare	Outpatient Healthcare	3.4%	99.6	88.2	11.4%	228.9	200.4	12.5%
Healthcare	Hospital	4.5%	100.4	91.5	8.9%	236.7	214.6	9.3%
Lodging	Small Hotel	1.6%	61.3	51.9	15.3%	119.0	99.5	16.4%
Lodging	Large Hotel	4.2%	84.4	73.5	12.9%	164.8	143.8	12.7%
Warehouse	Warehouse	18.6%	13.8	10.6	23.2%	27.1	18.6	31.4%
Food Service	Quick-Service Restaurant	0.3%	502.2	469.4	6.5%	860.8	808.9	6.0%
Food Service	Full-Service Restaurant	1.0%	341.5	316.5	7.3%	641.8	600.0	6.5%
Apartment	Mid-Rise Apartment	13.7%	39.3	33.6	14.5%	103.6	88.7	14.4%
Apartment	High-Rise Apartment	9.6%	45.3	39.0	13.9%	95.3	82.4	13.5%
National	Weighted Average	100%	47.8	41.1	14.0%	108.5	92.5	14.7%

Site EUI units: kBtu/ft²-yr. Source EUI units: kBtu/ft²-yr.

3.3 Climate-Zone Comparison (Net Energy, including on-site renewables)

Climate zone	Weight	2019 site	2022 site	Site savings	2019 source	2022 source	Source savings
1A	0.04	46.5	40.6	12.7%	115.6	100.2	13.3%
2A	0.17	45.3	39.2	13.5%	113.1	96.8	14.4%
2B	0.03	41.2	35.2	14.6%	103.6	87.5	15.5%
3A	0.15	45.8	39.6	13.5%	107.4	91.6	14.7%
3B	0.09	39.4	33.4	15.2%	95.3	79.9	16.2%
3C	0.02	39.0	33.3	14.6%	96.9	81.6	15.8%
4A	0.21	48.2	41.4	14.1%	106.2	90.8	14.5%
4B	0.00	49.4	42.1	14.8%	113.0	95.5	15.5%
4C	0.03	41.0	35.4	13.7%	93.5	80.4	14.0%
5A	0.18	55.0	47.1	14.4%	113.0	96.3	14.8%
5B	0.05	48.8	41.7	14.5%	107.8	91.5	15.1%
5C	0.00	54.4	46.7	14.2%	116.9	100.9	13.7%
6A	0.03	64.8	55.4	14.5%	129.5	110.2	14.9%
6B	0.01	60.5	51.7	14.5%	123.5	105.6	14.5%
7	0.00	70.2	60.7	13.5%	137.5	118.8	13.6%
8	0.00	87.5	75.7	13.5%	156.4	136.0	13.0%
National	1.00	47.8	41.1	14.0%	108.5	92.5	14.7%

Site EUI units: kBtu/ft2-yr. Source EUI units: kBtu/ft2-yr.

3.4 Gross Energy Use Intensity by Climate Zone -- Standard 90.1-2022 (excluding renewables)

Climate Zone	Floor Area Weight	Site EUI (kBtu/ft2)	Site EUI (kWh/m2)	Source EUI (kBtu/ft2)	ECI (\$/ft2-yr)	CO2 (tons/kft2)
1A	0.04	42.5	134.1	105.5	\$1.34	7.5
2A	0.17	41.4	130.6	103.1	\$1.31	7.3
2B	0.03	37.7	118.9	94.6	\$1.20	6.7
3A	0.15	41.8	131.8	97.8	\$1.23	6.8
3B	0.09	36.0	113.6	87.2	\$1.10	6.1
3C	0.02	35.5	112.0	87.9	\$1.11	6.2
4A	0.21	43.2	136.2	95.8	\$1.19	6.6
4B	0.00	44.8	141.3	103.0	\$1.29	7.2
4C	0.03	36.9	116.4	84.6	\$1.06	5.9
5A	0.18	48.9	154.3	101.4	\$1.24	6.8
5B	0.05	43.8	138.1	97.6	\$1.21	6.7
5C	0.00	48.5	153.0	105.9	\$1.31	7.2
6A	0.03	57.2	180.4	115.5	\$1.41	7.7
6B	0.01	53.6	169.1	111.2	\$1.36	7.5
7	0.00	62.5	197.2	123.8	\$1.50	8.2
8	0.00	77.1	243.2	140.0	\$1.67	9.0
National	1.00	43.1	135.9	98.3	\$1.23	6.8

3.5 Gross Energy Use Intensity by Building Type -- Standard 90.1-2022 (excluding renewables)

Building Type	Prototype	Floor Area Weight	Site EUI (kBtu/ft2)	Site EUI (kWh/m2)	Source EUI (kBtu/ft2)	ECI (\$/ft2-yr)
Office	Small Office	3.8%	25.7	81.1	71.4	\$0.92
Office	Medium Office	5.0%	27.5	86.7	70.4	\$0.90
Office	Large Office	3.9%	50.2	158.4	137.5	\$1.77
Retail	Standalone Retail	10.9%	39.3	123.9	87.9	\$1.09
Retail	Strip Mall	3.7%	40.6	128.1	99.7	\$1.26
Education	Primary School	4.8%	41.6	131.2	97.0	\$1.22
Education	Secondary School	10.9%	36.4	114.8	87.7	\$1.11
Healthcare	Outpatient	3.4%	90.8	286.3	207.8	\$2.60
Healthcare	Hospital	4.5%	93.0	293.3	218.9	\$2.75
Lodging	Small Hotel	1.6%	53.9	170.0	105.2	\$1.27
Lodging	Large Hotel	4.2%	75.0	236.6	148.1	\$1.80
Warehouse	Warehouse	18.6%	12.8	40.4	25.0	\$0.30
Food Service	Quick-Service Restaurant	0.3%	469.4	1,480.8	808.9	\$9.50
Food Service	Full-Service Restaurant	1.0%	316.5	998.5	600.0	\$7.21
Apartment	Mid-Rise Apartment	13.7%	35.5	112.0	94.2	\$1.21
Apartment	High-Rise Apartment	9.6%	40.0	126.2	85.3	\$1.05
National	Weighted Average	100%	43.1	135.9	98.3	\$1.23

3.6 Percent Gross Energy Savings by Climate Zone (2022 vs 2019)

Climate Zone	Floor Area Weight	Site EUI (%)	Source EUI (%)	ECI (%)	CO2 (%)
1A	0.04	8.6	8.7	8.8	8.5
2A	0.17	8.6	8.8	8.4	8.8
2B	0.03	8.5	8.7	9.1	9.5
3A	0.15	8.7	8.9	8.9	9.3
3B	0.09	8.6	8.5	8.3	9.0
3C	0.02	9.0	9.3	9.8	10.1
4A	0.21	10.4	9.8	9.8	9.6
4B	0.00	9.3	8.8	8.5	7.7
4C	0.03	10.0	9.5	9.4	9.2
5A	0.18	11.1	10.3	10.1	10.5
5B	0.05	10.2	9.5	9.7	9.5
5C	0.00	10.8	9.4	9.0	10.0
6A	0.03	11.7	10.8	10.2	10.5
6B	0.01	11.4	10.0	9.9	9.6
7	0.00	11.0	10.0	10.2	9.9
8	0.00	11.9	10.5	9.7	10.0
National	1.00	9.8	9.4	8.9	9.3

3.7 Practical Reading Notes

- The national weighted average improvement is material: 14.0% net site-EUI savings and 14.7% net source-EUI savings from 90.1-2019 to 90.1-2022.
- Net savings are higher than gross savings because 90.1-2022 introduced mandatory on-site renewable energy requirements (Section 11 / Addendum ap).
- In the building-type results, Warehouse and Strip Mall show the strongest source-EUI savings.
- In the climate-zone results, climate zone 3B shows the strongest source-EUI savings at 16.2%.
- CZ6A gross site EUI: 57.2 kBtu/ft² (180.4 kWh/m²); CZ6A net site EUI: 55.4 kBtu/ft² (174.8 kWh/m²).

4. Energy Savings Analysis: 2024 IECC for Residential Buildings (December 2024)

Item	Value
Full title	Energy Savings Analysis: 2024 IECC for Residential Buildings
Publication date	December 2024
Coverage	U.S. residential buildings
Model family	32 prototype configurations: 2 building types x 4 foundation types x 4 fuel/equipment types; simulated across 18 climate/moisture regimes
Simulation engine	EnergyPlus
Main comparison	2021 IECC vs 2024 IECC
Interpretation	This is a weighted residential code-savings study, not a measured stock median dataset.

4.1 National Weighted Summary

Version	Site EUI (kBtu/ft2-yr)	Source EUI (kBtu/ft2-yr)	Energy Cost (\$/residence-yr)	CO2 (tons/residence-yr)
2021 IECC	31.3	68.5	2,526	10.84
2024 IECC	28.9	63.9	2,360	10.13
Savings	7.80%	6.80%	6.60%	6.51%

4.2 Climate-Zone Comparison

Climate zone	Weight	2021 site	2024 site	Site savings	2021 source	2024 source	Source savings	Cost savings	CO2 savings
1	2.12	26.7	24.8	7.07%	69.4	63.9	7.95%	8.09%	8.15%
2	26.02	27.0	24.9	7.90%	67.4	62.0	7.96%	7.97%	7.97%
3	28.84	26.4	24.7	6.38%	63.9	60.0	6.13%	6.08%	6.06%
4	19.07	31.3	28.9	7.43%	68.1	63.4	6.90%	6.79%	6.74%
5	18.33	38.7	35.9	7.31%	71.7	67.5	5.88%	5.53%	5.36%
6	5.05	47.0	40.8	13.23%	83.6	77.6	7.20%	5.64%	4.87%
7	0.55	52.0	44.8	13.84%	93.1	84.0	9.79%	8.74%	8.23%
8	0.01	65.8	56.4	14.29%	111.8	99.8	10.73%	9.74%	9.26%
National	100.00	31.3	28.9	7.80%	68.5	63.9	6.80%	6.60%	6.51%

Site EUI and Source EUI units: kBtu/ft2-yr.

4.3 Building-Type Comparison

Building type	Weight	2021 site	2024 site	Site savings	2021 source	2024 source	Source savings	Cost savings	CO2 savings
Single-family	82.12	30.9	28.4	7.94%	67.6	63.0	6.77%	6.54%	6.43%
Multifamily Unit	17.88	34.9	32.6	6.60%	76.9	71.5	7.05%	7.14%	7.18%
National	100.00	31.3	28.9	7.80%	68.5	63.9	6.80%	6.60%	6.51%

4.4 CZ6 Residential EUI in kWh/m2

Metric	2021 IECC	2024 IECC	Savings
Site EUI	47.0 kBtu/ft2 (148.3 kWh/m2)	40.8 kBtu/ft2 (128.7 kWh/m2)	13.23%
Source EUI	83.6 kBtu/ft2 (263.6 kWh/m2)	77.6 kBtu/ft2 (244.8 kWh/m2)	7.20%
Energy Cost	\$3,231/yr	\$3,049/yr	5.64%

4.5 Weighting Factors

Category	Subcategory	Weight (%)
Building Type	Single-family	82.12
	Multifamily Unit	17.88
Heating System	Gas-Fired Furnace	55.60
	Electric Furnace	7.88
	Oil-Fired Furnace	0.15
	Heat Pump	36.37
Climate Zone	CZ1	2.12
	CZ2	26.02
	CZ3	28.84
	CZ4	19.07
	CZ5	18.33
	CZ6	5.05
	CZ7	0.55
	CZ8	0.01

4.6 Practical Reading Notes

- The national result is more modest than the 90.1-2022 commercial delta: 7.80% site-EUI savings and 6.80% source-EUI savings versus 2021 IECC.

- Stronger site-EUI savings appear in colder climate zones 6 through 8, with climate zone 8 showing the highest site-EUI savings at 14.29%.
- Multifamily shows lower site-EUI savings than single-family, but slightly higher source-EUI, cost, and CO2 savings.
- CZ6 shows the largest site-EUI savings (13.23%) among regularly weighted climate zones -- this is the most relevant zone for this project's Canadian/cold-climate context.

Appendix A: Cross-Reference Summary for BEM Validation

Residential Buildings (CZ6, Cold Climate)

Source	Standard/Year	Site EUI (kBtu/ft2)	Site EUI (kWh/m2)	Notes
IECC 2024	2024 IECC, CZ6	40.8	128.7	New construction, weighted avg
IECC 2021	2021 IECC, CZ6	47.0	148.3	New construction, weighted avg
IECC 2024	National, single-family	28.4	89.6	All CZs weighted
IECC 2024	National, multifamily	32.6	102.8	All CZs weighted

Commercial Buildings (CZ6A, Cold Climate)

Source	Standard/Year	Site EUI (kBtu/ft2)	Site EUI (kWh/m2)	Notes
ASHRAE 90.1-2022	CZ6A Gross	57.2	180.4	Weighted avg, all 16 prototypes
ASHRAE 90.1-2022	CZ6A Net	55.4	174.8	Including on-site renewables
ASHRAE 90.1-2019	CZ6A	64.8	204.4	Baseline
DOE 2008	CZ6A new construction	34.6-115.2	109-363	Range across building types

Selected Building Types -- Multi-Source CZ6A Comparison

Building Type	ASHRAE 90.1-2022 Gross (kWh/m2)	DOE 2008 CZ6A (kWh/m2)	NRCan Median (kWh/m2)
Small Office	81.1	109.1	430.6
Mid-Rise Apartment	112.0	--	N/A
High-Rise Apartment	126.2	--	N/A
Standalone Retail	123.9	109.8	444.4
Primary School	131.2	363.4	297.2
Hospital	293.3	486.4	858.3
Small Hotel	170.0	230.9	380.6
Warehouse	40.4	168.1	283.3

Key Observations:

1. ASHRAE 90.1-2022 values represent code-minimum new construction; NRCan values represent the median of all existing building stock (all ages). The large gap is expected.
 2. DOE 2008 benchmarks used Standard 90.1-2004 requirements, which are significantly less stringent than 90.1-2022, explaining their higher values.
 3. For CZ6 residential validation: simulation EUI values in the 128-140 kWh/m² range align well with the 2024 IECC CZ6 reference of 128.7 kWh/m².
 4. Do not compare all four sources as if they were the same kind of number. DOE and ASHRAE/IECC are simulation-based studies; NRCan is a benchmark median sheet from survey data.
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Practical Use Notes for This Repo

- Use **DOE 2008** when you need the older benchmark-model archetypes, geometry scale, HVAC system types, and broad climate sensitivity for commercial buildings.
 - Use **NRCan 2018** for benchmarking against Canadian medians and for checking whether simulated outputs are broadly plausible in a Canadian context.
 - Use **ASHRAE 90.1-2022** for commercial code-delta expectations between recent code editions and for CZ6A EUI validation targets.
 - Use **2024 IECC** for residential code-delta expectations between 2021 and 2024, especially the CZ6 reference of 128.7 kWh/m².
 - Do not compare all four sources as if they were the same kind of number. DOE and ASHRAE/IECC are simulation-based studies; NRCan is a benchmark median sheet.
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